Reading in Data

The first step is to read in the Crowd Counting Consortium data. I downloaded the three available phases; 2017-2020, 2021-2024, and 2025's currently availiable data.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(lubridate)
```

```
ccc_files <- list.files("data/raw/ccc", pattern = "\\.csv$", full.names = TRUE)
ccc_files
```

```
[1] "data/raw/ccc/ccc_compiled_20172020.csv"
[2] "data/raw/ccc/ccc_compiled_20212024.csv"
[3] "data/raw/ccc/ccc-phase3-public.csv"
```

I want to stack the three csv files into one table.

```r
ccc_raw <- ccc_files |>
  map(\(f) read_csv(f, col_types = cols(.default = col_character())) |>
        rename_with(tolower)) |>
   bind_rows()

names(ccc_raw)
```

```
  [1] "date"                    "locality"
  [3] "state"                   "location_detail"
  [5] "online"                  "type"
  [7] "macroevent"              "actors"
  [9] "claims"                  "valence"
 [11] "issues"                  "size_text"
 [13] "size_low"                "size_high"
 [15] "size_mean"               "size_cat"
 [17] "arrests"                 "arrests_any"
 [19] "injuries_crowd"          "injuries_crowd_any"
 [21] "injuries_police"         "injuries_police_any"
 [23] "property_damage"         "property_damage_any"
 [25] "chemical_agents"         "source_1"
 [27] "source_2"                "source_3"
 [29] "source_4"                "source_5"
 [31] "source_6"                "source_7"
 [33] "source_8"                "source_9"
 [35] "source_10"               "source_11"
 [37] "source_12"               "source_13"
 [39] "source_14"               "source_15"
 [41] "source_16"               "source_17"
 [43] "source_18"               "source_19"
 [45] "source_20"               "source_21"
 [47] "source_22"               "source_23"
 [49] "source_24"               "source_25"
 [51] "source_26"               "source_27"
 [53] "source_28"               "source_29"
 [55] "source_30"               "notes"
 [57] "lat"                     "lon"
 [59] "resolved_locality"       "resolved_county"
 [61] "resolved_state"          "fips_code"
 [63] "title"                   "organizations"
 [65] "participants"            "claims_summary"
 [67] "claims_verbatim"         "issue_tags_summary"
 [69] "issue_tags_verbatim"     "issue_tags"
```

```
 [71] "participant_measures"       "police_measures"
 [73] "participant_deaths"         "police_deaths"
 [75] "location"                   "event_type"
 [77] "notables"                   "targets"
 [79] "conf"                       "participant_injuries"
 [81] "police_injuries"            "participant_casualties_any"
 [83] "police_casualties_any"      "repeater"
 [85] "source1"                    "source2"
 [87] "source3"                    "source4"
 [89] "source5"                    "source6"
 [91] "source7"                    "source8"
 [93] "source9"                    "source10"
 [95] "source11"                   "source12"
 [97] "source13"                   "source14"
 [99] "source15"                   "source16"
[101] "source17"                   "source18"
[103] "source19"                   "source20"
[105] "source21"                   "source22"
[107] "source23"                   "source24"
[109] "source25"                   "source26"
[111] "source27"                   "source28"
[113] "source29"                   "source30"
```

```
nrow
```

```
function (x)
dim(x)[1L]
<bytecode: 0x11a813ff8>
<environment: namespace:base>
```

```
ccc_raw |>
  head()
```

```
# A tibble: 6 x 114
  date      locality state location_detail online type  macroevent actors claims
  <chr>     <chr>    <chr> <chr>           <chr>  <chr> <chr>      <chr>  <chr>
1 2017-01-~ Washing~ DC    Lafayette Squa~ 0      vigil <NA>       <NA>   for b~
2 2017-01-~ Mankato  MN    <NA>            0      vigil <NA>       Peace~ for p~
3 2017-01-~ Minneap~ MN    U.S. Bank Stad~ 0      prot~ <NA>       gener~ again~
4 2017-01-~ Little ~ RI    Town Green      0      vigil <NA>       Sakon~ for p~
5 2017-01-~ Oak Rid~ TN    Y-12 National ~ 0      vigil <NA>       Oak R~ for a~
```

```
6 2017-01-~ Washing~ DC    Lafayette Squa~ 0      vigil <NA>       <NA>   for b~
# i 105 more variables: valence <chr>, issues <chr>, size_text <chr>,
#   size_low <chr>, size_high <chr>, size_mean <chr>, size_cat <chr>,
#   arrests <chr>, arrests_any <chr>, injuries_crowd <chr>,
#   injuries_crowd_any <chr>, injuries_police <chr>, injuries_police_any <chr>,
#   property_damage <chr>, property_damage_any <chr>, chemical_agents <chr>,
#   source_1 <chr>, source_2 <chr>, source_3 <chr>, source_4 <chr>,
#   source_5 <chr>, source_6 <chr>, source_7 <chr>, source_8 <chr>, ...
```

I have run into a problem where the three phases all call the "protest inciting issue" different column names. The older phases use "issues," whereas the newer phases use "issues_tags."

```
ccc_raw |>
  summarise(
    issues       = sum(!is.na(issues)),
    issue_tags   = sum(!is.na(issue_tags)),
    tags_summary = sum(!is.na(issue_tags_summary)),
    total_rows   = n()
  )
```

```
# A tibble: 1 x 4
  issues issue_tags tags_summary total_rows
   <int>      <int>        <int>      <int>
1 131060     137452       198532     275151
```

```
ccc_raw <-
  ccc_raw |>
  mutate(topic = coalesce(issue_tags, issues, issue_tags_summary))

sum(is.na(ccc_raw$topic))
```

```
[1] 6639
```

```
ccc_raw |> count(topic, sort = TRUE) |> head(20)
```

```
# A tibble: 20 x 2
   topic                                                                  n
   <chr>                                                              <int>
 1 foreign affairs                                                    23772
 2 policing; racism                                                   11820
```

```
 3 military                                                                 11196
 4 policing;racism                                                           9671
 5 immigration                                                               7807
 6 education; guns                                                           7299
 7 labor                                                                     7008
 8 <NA>                                                                      6639
 9 democracy;presidency                                                      6163
10 presidency                                                                5757
11 environment                                                               5463
12 racism                                                                    4242
13 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
14 religion;reproductive rights                                              3513
15 guns                                                                      3048
16 democracy                                                                 2962
17 healthcare;labor                                                          2948
18 foreign affairs;military                                                  2781
19 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
20 lgbtqia                                                                   2466
```

```
sum(is.na(ccc_raw$topic))
```

```
[1] 6639
```

```
climate <- ccc_raw |>
  filter(str_detect(str_to_lower(topic), "environment|climate"))

nrow(climate)
```

```
[1] 23773
```

```
climate |> count(topic, sort = TRUE) |> head(15)
```

```
# A tibble: 15 x 2
   topic                                                                    n
   <chr>                                                                <int>
 1 environment                                                           5463
 2 democracy;economy;education;environment;guns;healthcare;immigration;pr~  3564
 3 corruption;democracy;economy;education;environment;guns;healthcare;imm~  2507
 4 democracy;economy;environment;healthcare;housing;immigration;labor and~   836
 5 development;environment;policing;racism                                741
```

```
 6 banking and finance;energy;environment                                    627
 7 energy;environment                                                        540
 8 energy;environment;indigenous peoples                                     361
 9 banking and finance;economy;energy;environment                            360
10 development;energy;environment                                           359
11 energy; environment                                                      352
12 civil rights; disability rights; economy; environment; immigration; la~   332
13 development; economy; energy; environment; indigenous peoples; labor; ~   293
14 environment;indigenous peoples                                           225
15 civil rights;economy;environment;guns;immigration;labor and workers ri~   219
```

```
ccc_raw |>
  filter(is.na(topic)) |>
  mutate(year = str_sub(date, 1, 4)) |>
  count(year)
```

```
# A tibble: 10 x 2
   year      n
   <chr> <int>
 1 2017   1081
 2 2018   1530
 3 2019    567
 4 2020    409
 5 2021    594
 6 2022    616
 7 2023    545
 8 2024    616
 9 2025    442
10 2026    239
```

Next, I will turn each protest date into a month and count how many climate and environmental protests happened in each month. This will allow me to find the protest waves (a month in the top 10% of monthly counts), which will tell me what wikipedia data I need to download.

```
climate <-
  climate |>
  mutate(
    date  = as_date(parse_date_time(date, orders = c("ymd", "mdy"))),
    month = floor_date(date, "month")
  )

nrow(climate)
```

```
[1] 23773
```

```
range(climate$date, na.rm = TRUE)
```

```
[1] "2017-01-01" "2026-05-31"
```

```
sum(is.na(climate$date))
```

```
[1] 0
```

```
climate <- climate |>
  mutate(
    date  = as_date(parse_date_time(date, orders = c("ymd", "mdy"))),
    month = floor_date(date, "month")
  )

nrow(climate)
```

```
[1] 23773
```

```
range(climate$date, na.rm = TRUE)
```

```
[1] "2017-01-01" "2026-05-31"
```

```
sum(is.na(climate$date))
```

```
[1] 0
```

```
monthly <-
  climate |>
  count(month, name = "protests")
monthly |> arrange(month)
```

```
# A tibble: 113 x 2
   month      protests
   <date>        <int>
 1 2017-01-01       29
 2 2017-02-01       58
```

```
 3 2017-03-01       49
 4 2017-04-01      123
 5 2017-05-01       22
 6 2017-06-01       38
 7 2017-07-01       28
 8 2017-08-01       20
 9 2017-09-01       19
10 2017-10-01       27
# i 103 more rows
```

```
head(10)
```

```
[1] 10
```

```
exists("climate")
```

```
[1] TRUE
```